

CSL7870: TASK-10 — Experiments on Karate Club and Journaling the Observations

Anuj Rajan Lalla (B22AI061)

September 11, 2025

Task Brief

Goal. Visualize and quantify node-wise receptive fields (message-passing trees) on the Karate Club graph and study how they grow with depth.

Graph. Zachary’s Karate Club (34 nodes).

Trees. Cycle-free BFS trees rooted at each node (depth k).

Setup and Visualization Choices

- **Extraction.** For a root v , build the depth- k BFS tree $T_k(v)$ (levels: L_0 root, L_1 neighbors, L_2 neighbors-of-neighbors, $\dots$), and a depth- k *induced* subgraph for metrics.
- **Coloring by class (why).** Nodes are colored by their Karate *club* label to **immediately see in how many hops a node reaches the opposite class**, which motivates the class-aware measurements (nearest-opposite hop, boundary index, 2-hop opposite fraction).

Metric Glossary (used in tables)

- **Tnodes** — size of depth- k BFS tree (receptive field).
- **induced_nodes** — node count in depth- k induced subgraph.
- **removed_edges** = edges(induced) – edges(tree): non-tree cross-edges/cycles pruned by BFS.
- **triangles** — # of 3-cycles in the induced subgraph; **avg_clust** — its average clustering.
- **deg** = $|L_1|$; **opp1/opp2** — opposite-class counts in L_1/L_2 ; **bndry_idx** = opp1/deg.
- **L1_opp/L2_opp** — opposite-class fraction on ring L_1/L_2 ; **minhop_opp** — first hop where opposite appears.
- **Gateways** — L1 intermediates (mid:count) carrying opposite-class 2-hop links.
- **coverage** — fraction of nodes reachable within k hops.

Experiment 1 — Depth-2 Trees for All 34 Roots (Tables)

Table 1: Depth-2 BFS Metrics (per node)

node	Tnodes	induced	removed	triangles	avg_clust
0	26	26	34	34	0.506
1	23	23	31	32	0.491
2	31	31	41	42	0.570
3	23	23	31	32	0.491
4	18	18	19	26	0.680
5	18	18	19	26	0.680
6	18	18	19	26	0.680
7	22	22	23	27	0.510
8	31	31	41	43	0.601
9	23	23	34	36	0.591
10	18	18	19	26	0.680
11	17	17	18	25	0.681
12	17	17	18	25	0.681
13	31	31	41	43	0.601
14	19	19	20	18	0.564
15	19	19	20	18	0.564
16	6	6	5	5	0.667
17	18	18	19	25	0.601
18	19	19	20	18	0.564
19	31	31	41	43	0.601
20	19	19	20	18	0.564
21	18	18	19	25	0.601
22	19	19	20	18	0.564
23	21	21	23	19	0.504
24	10	10	10	5	0.243
25	10	10	9	8	0.467
26	18	18	15	17	0.685
27	25	25	37	37	0.547
28	25	25	37	37	0.547
29	20	20	21	18	0.505
30	25	25	36	38	0.614
31	33	33	44	44	0.568
32	25	25	37	37	0.547
33	24	24	34	31	0.531

Table 2: Class-Aware Hop Stats (per node)

node	club	deg	opp1	opp2	bndry_idx	L1_opp	L2_opp	minhop_opp
0	Mr. Hi	16	1	8	0.06	0.06	0.89	1
1	Mr. Hi	9	1	6	0.11	0.11	0.46	1
2	Mr. Hi	10	4	11	0.40	0.40	0.55	1
3	Mr. Hi	6	0	7	0.00	0.00	0.44	2
4	Mr. Hi	3	0	1	0.00	0.00	0.07	2
5	Mr. Hi	4	0	1	0.00	0.00	0.08	2
6	Mr. Hi	4	0	1	0.00	0.00	0.08	2
7	Mr. Hi	4	0	6	0.00	0.00	0.35	2
8	Mr. Hi	5	3	12	0.60	0.60	0.48	1
9	Officer	2	1	7	0.50	0.50	0.35	1

10	Mr. Hi	3	0	1	0.00	0.00	0.07	2
11	Mr. Hi	1	0	1	0.00	0.00	0.07	2
12	Mr. Hi	2	0	1	0.00	0.00	0.07	2
13	Mr. Hi	5	1	14	0.20	0.20	0.56	1
14	Officer	2	0	4	0.00	0.00	0.25	2
15	Officer	2	0	4	0.00	0.00	0.25	2
16	Mr. Hi	2	0	0	0.00	0.00	0.00	3
17	Mr. Hi	2	0	2	0.00	0.00	0.13	2
18	Officer	2	0	4	0.00	0.00	0.25	2
19	Mr. Hi	3	1	14	0.33	0.33	0.52	1
20	Officer	2	0	4	0.00	0.00	0.25	2
21	Mr. Hi	2	0	2	0.00	0.00	0.13	2
22	Officer	2	0	4	0.00	0.00	0.25	2
23	Officer	5	0	4	0.00	0.00	0.27	2
24	Officer	3	0	2	0.00	0.00	0.33	2
25	Officer	3	0	1	0.00	0.00	0.17	2
26	Officer	2	0	3	0.00	0.00	0.20	2
27	Officer	4	1	7	0.25	0.25	0.35	1
28	Officer	3	1	7	0.33	0.33	0.33	1
29	Officer	4	0	4	0.00	0.00	0.27	2
30	Officer	4	2	8	0.50	0.50	0.40	1
31	Officer	6	1	15	0.17	0.17	0.58	1
32	Officer	12	2	6	0.17	0.17	0.50	1
33	Officer	17	3	4	0.18	0.18	0.67	1

Table 3: Gateways to the Opposite Class (per node)

node	total_opp_2hop	top_gateways (mid:count)
0	8	2:4, 31:2, 1:1
1	6	2:4, 0:1, 13:1
2	11	32:6, 27:2, 0:1
3	7	2:4, 0:1, 1:1
4	1	0:1
5	1	0:1
6	1	0:1
7	6	2:4, 0:1, 1:1
8	12	32:7, 2:3, 0:1
9	7	2:6, 33:1
10	1	0:1
11	1	0:1
12	1	0:1
13	14	33:8, 2:4, 0:1
14	4	32:2, 33:2
15	4	32:2, 33:2
16	0	-
17	2	0:1, 1:1
18	4	32:2, 33:2
19	14	33:12, 0:1, 1:1
20	4	32:2, 33:2
21	2	0:1, 1:1
22	4	32:2, 33:2
23	4	33:2, 27:1, 32:1
24	2	27:1, 31:1
25	1	31:1
26	3	33:3
27	7	2:6, 33:1
28	7	2:6, 33:1
29	4	32:2, 33:2
30	8	1:8
31	15	0:15
32	6	2:5, 33:1
33	4	8:2, 13:2

Table 4: Class-Level Means

club	mean_boundary_idx	mean_minhop_opp	mean_L2_opp_frac
Mr. Hi	0.10	1.71	0.29
Officer	0.12	1.59	0.33

Table 5: Same-Class Pairs with Highest Overlap (Opposite 2-Hop Sets)

Top pairs (Jaccard)
(4,5) J=1.00; (4,6) J=1.00; (4,10) J=1.00; (4,11) J=1.00; (4,12) J=1.00; (5,6) J=1.00; (5,10) J=1.00; (5,11) J=1.00

Experiment 2 — Depth Growth (3 vs. 4 hops) — Tables

root	depth	tree_nodes	induced_nodes	coverage	removed_edges	triangles
24	3	33	33	0.97	44	44
24	4	34	34	1.00	45	45
26	3	24	24	0.71	34	31
26	4	33	33	0.97	44	44
2	3	34	34	1.00	45	45
2	4	34	34	1.00	45	45

Interpretations

Experiment 1 — Depth-2 message-passing trees

Global mixing at two hops. From the hop table, opposite-class nodes are reachable within two hops for **33/34** roots (13 at 1 hop, 20 at 2 hops; only node 16 needs > 2). Practically, a **2-layer GNN** already mixes cross-class signals for almost every node.

Central vs. peripheral receptive fields. The depth-2 tree size (**Tnodes**) separates positions in the graph. **Large trees** (e.g., nodes 31: 33; 2/8/13/19: 31) indicate **central** roots that “see” most of the graph within two steps. **Small trees** (e.g., node 16: 6; nodes 24/25: 10) mark **peripheral** roots whose 2-hop field is localized.

Redundancy pruned by BFS. Around central roots the induced 2-hop subgraph contains many non-tree edges and triangles; the BFS view removes these to reveal layered flow. This is reflected by **removed_edges** and **triangles**, which peak near hubs (e.g., node 31: 44/44; nodes 2/8/13/19: ≈ 41 –43). Interpretation: central regions have **many alternative routes**, so message passing will *rapidly mix* information there.

Boundary strength and nearest-opposite hop. The class-aware table shows which roots lie on the community cut. **Hard-boundary** nodes (min-hop = 1 with high **bndry_idx**, e.g., nodes 8: 0.60; 30: 0.50; 2: 0.40; 9: 0.50) already touch the opposite class at one hop and will blur most quickly under 2+ layers. **Soft-boundary** leaves (e.g., 4–7, 10–12, 17, 21) have zero opposite at L_1 but nonzero at L_2 (min-hop = 2). Node 16 (min-hop = 3) is deep interior at this scale.

Gateways (who carries cross-class signal). Opposite-class 2-hop links concentrate through a few level-1 intermediates: e.g., **31** $\rightarrow$ **0** (15 links), **30** $\rightarrow$ **1** (8), **2/8** $\rightarrow$ **32** (6–7), **13/19** $\rightarrow$ **33** (8–12). These are **dominant corridors** of cross-class flow; ablating or down-weighting them would markedly reduce mixing.

Class-level asymmetry. Club-wise means show **Officers** are slightly more exposed: mean nearest-opposite hop **1.59** vs. **1.71** (Mr. Hi), and mean L_2 opposite fraction **0.33** vs. **0.29**. Hence the Officer side tends to mix a bit earlier by depth 2.

Structural equivalence at the leaves. Several Mr. Hi leaves (e.g., {4, 5, 6, 10, 11, 12}) have **identical** opposite 2-hop sets (Jaccard = 1.0 across many pairs). Under two layers of message passing they will become **indistinguishable** unless distinguished by features or positional/structural encodings.

Exp-1 takeaway. Two hops already expose nearly everyone to the other class; central nodes have large, highly redundant fields and will mix quickest; a handful of gateways (0/1/32/33) dominate cross-class transit; a pocket of leaves are structurally equivalent at this depth.

Experiment 2 — Depth growth: 3 vs. 4 hops

Root 24 (Officer). Coverage rises from **97%** (33/34) at 3 hops to **100%** at 4; the extra hop mainly increases `removed_edges/triangles` (44→45), i.e., adds **redundant cycles** rather than new nodes. Near-saturation by 3 hops.

Root 26 (Officer). Coverage jumps from **71%** (24/34) to **97%** (33/34) when going 3→4 hops, with a marked rise in redundancy (removed 34→44, triangles 31→44). A more **peripheral** root that **requires the 4th hop** to reach most of the graph.

Root 2 (Mr. Hi). Already **100%** coverage at 3 hops; depth 4 brings no new nodes (only maximal redundancy: 45/45). Highly **central**; **3 layers suffice** and deeper stacks likely oversmooth.

Exp-2 takeaway. Mixing speed is **heterogeneous**: some roots are global by 3 hops, others only by 4. Once coverage $\gtrsim 97\%$, deeper hops add mainly **cycles**, not **new evidence**.